Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Predictive Modeling, Incident Volume Prediction, Capacity Planning, Workforce Demand Modeling, Severity Classification, Intelligent Triaging, Anomaly Detection, Time-Series Forecasting (SARIMA), Dimensional Data Modeling, Star/Snowflake Schemas, Slowly Changing Dimensions (SCD Type 1/2), Statistical Modeling, Experimental Design, Data Quality Auditing, Agentic AI, Generative AI, Root Cause Analysis (RCA)

Programming Languages & Tools: Python, SQL, R, XGBoost, Random Forests, Scikit-Learn, SARIMA, AWS (S3, Redshift, AWS Glue, Lambda), Azure Cloud (ADF, Azure Synapse, Azure SQL, Databricks), Snowflake, Apache Spark, PySpark, Power BI, Tableau, MicroStrategy, Airflow, One Automation, KIRO IDE

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data Scientist, IT Operations & Capacity Planning — **May 2022 – Present**

- **Architected end-to-end predictive machine learning models** using XGBoost, Random Forests, and Scikit-Learn to forecast IT service demand and capacity planning, optimizing workforce capacity and reducing operational response times by 20% at Charter Communications.
- **Designed advanced time-series forecasting frameworks (SARIMA)** to predict customer incident volume and truck rolls, driving proactive resource allocation and establishing robust data science solutions for operational excellence.
- **Engineered proactive anomaly detection and classification systems** utilizing machine learning algorithms with 95% accuracy to process high-throughput IT service records, identify emerging operational patterns, and mitigate discrepancies.
- **Built automated data quality monitoring and audit frameworks** featuring operational firewalls and scalable data pipelines in AWS (S3, Redshift) and Azure Cloud to ensure data integrity and reduce operational turnaround times by over 80%.
- **Pioneered agentic AI and automated QA testing frameworks** leveraging prompt engineering, Python, and PySpark to accelerate development delivery by 40% and support modern intelligent triaging models for operational infrastructure.

Graduate Research & Information Systems Analyst — **May 2022 – Present**

- **Developed dimensional data models and star schemas** leveraging high-performance SQL and relational database systems to streamline analytical reporting and centralize IT operational metrics.
- **Implemented robust data pipelines and statistical validation frameworks** utilizing advanced data science methodologies to optimize information delivery and ensure compliance with strict data availability SLAs.
- **Conducted rigorous exploratory data analysis and statistical modeling** using Python, R, and advanced analytics techniques to translate complex business problems into actionable, data-driven strategies.

Petrochemical Engineer & Operations Analyst — **May 2022 – Present**

- **Optimized industrial process workflows and capacity planning** by applying rigorous mathematical and statistical frameworks to analyze operational efficiency and resource utilization.
- **Designed cross-functional operational reports and metric dashboards** to monitor Key Operating Metrics (KOM), translating complex technical data into clear insights for senior stakeholders.
- **Spearheaded root cause analysis and process failure investigations** applying systematic troubleshooting methodologies to improve system reliability, safety compliance, and operational uptime.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture **Pune, India**

Test Engineering Analyst **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.